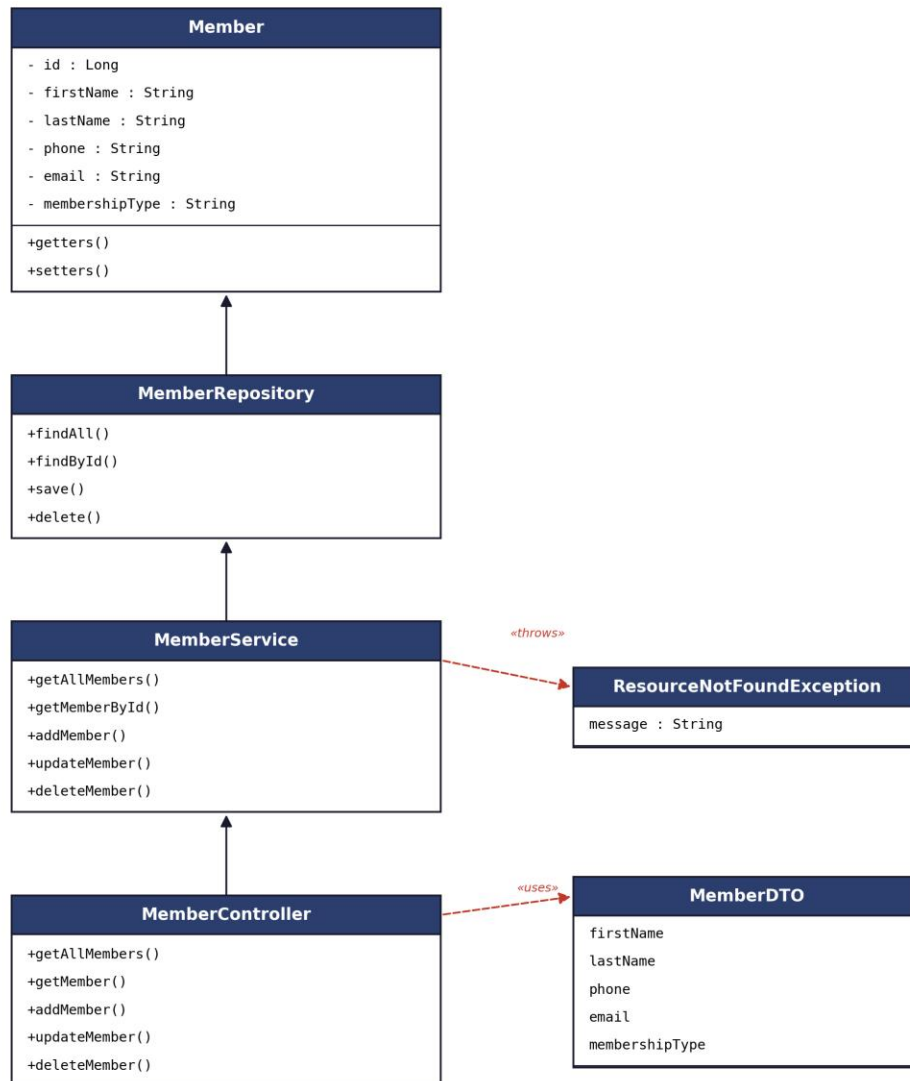


# Class Diagram: Member Management Module

*Member • MemberRepository • MemberService • MemberController • MemberDTO • ResourceNotFoundException*

**Class Diagram: Member Management Module**



## Class Descriptions

| Class         | Responsibility                                                                                                                                                                                                                                 |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Member</b> | Entity/model representing a gym or club member, with fields <code>id</code> , <code>firstName</code> , <code>lastName</code> , <code>phone</code> , <code>email</code> , and <code>membershipType</code> , plus generated getters and setters. |

|                                  |                                                                                                                                                                                  |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MemberRepository</b>          | Data access layer. Provides findAll(), findById(), save(), and delete() to persist and retrieve Member records.                                                                  |
| <b>MemberService</b>             | Business logic layer. Exposes getAllMembers(), getMemberById(), addMember(), updateMember(), and deleteMember(), calling into MemberRepository.                                  |
| <b>MemberController</b>          | API/presentation layer. Exposes REST endpoints (getAllMembers(), getMember(), addMember(), updateMember(), deleteMember()) that delegate to MemberService.                       |
| <b>MemberDTO</b>                 | Data Transfer Object used by the Controller to move firstName, lastName, phone, email, and membershipType between the API layer and the client without exposing the full entity. |
| <b>ResourceNotFoundException</b> | Custom exception carrying a message, thrown by the Service layer when a requested Member cannot be found.                                                                        |

## Relationships

MemberController depends on MemberService.

MemberService depends on MemberRepository.

MemberRepository depends on the Member entity.

MemberController uses MemberDTO to shape request/response data.

MemberService throws ResourceNotFoundException when a member lookup fails.